

The within-pair functor of a Fermat CRT block

A symmetric-square law for the within-pair survivor count,
its proof for odd characteristic, and the localization of the
characteristic-two obstruction to the generic swap torsion

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Abstract

This note determines the within-pair survivor count of an all- A pair block in the Chinese-Remainder splitting of a Fermat cell, the combinatorial object that controls the $\mathbb{F}_p$ -isomorphism question on the coupled flank of the author's computational campaign. Three results are established. First, the survivor count factorizes as (within-class) $\times$ (across-class) with the within-class factor always free (c^2) and the across-class factor either the full square $(p^v - 1)^2$ (the *free* regime) or the triangle $T(p^v - 1) = (p^v - 1)p^v/2$ (the *coupled* regime); the resulting quotient law coupled/free = $p^v/(2(p^v - 1))$ is derived, not fitted, and its degeneration coupled = free occurs exactly at $p^v = 2$. Second, the coupled object is identified functorially: with $V = \mathbb{F}_p^{p^v-1}$ the live-class space, the coupled survivor is $\dim(\mathrm{Sym}^2 V) \cdot c^2$ and the free survivor is $\dim(V \otimes V) \cdot c^2$, so the entire free/coupled distinction is the collapse of the tensor square to the symmetric square on the class layer; the triangular index set is exactly the canonical basis of $\mathrm{Sym}^2 V$, which resolves why the triangle survives a full-square support. Third, the realization question — does the relevant symmetrization map surject onto $\mathrm{Sym}^2 V$ — is answered: it is surjective in odd characteristic (so the coupled identity is a theorem there), and in characteristic two it falls short by exactly $p^v - 1$ dimensions, the cokernel being $(\mathbb{Z}/2)^{p^v-1}$. This cokernel is shown to be the *generic* torsion of the swap involution on $V \otimes V$, depending only on $\dim V$ and not on the Fermat data; hence the within-pair class layer is closed as a site for a Fermat-specific torsion counterexample. Every numerical statement is reproduced byte-exact by a self-contained verifier across the cells named in the text. The torsion counterexample has not been found; this note closes one candidate layer with data and proves the coupled law for odd characteristic.

1 Setting and the object

Fix a Fermat cell (n, m) with n even and a prime $p \mid m$, and write $m = p^v \cdot m'$ with $p \nmid m'$, so p^v is the p -part of m . Over $\mathbb{F}_p$ the relevant cyclotomic quotient factors into a $(t - 1)$ -primary part A and a coprime cofactor B ; the Chinese Remainder Theorem splits the per-variable ring as $A \times B$, and tensoring over the variables decomposes the cohomology module into CRT blocks indexed by which variables lie in A and which in B (the localization of torsion to a single such block is proven separately in [1]). This note concerns the interior of one block: the *all- A pair block*, in which two distinguished variables — a *pair* — both lie in the A -factor.

The single-variable signature is governed by the Frobenius identity $(t - 1)^{p^v} = t^{p^v} - 1$ over $\mathbb{F}_p$: the idempotent projecting onto the $(t - 1)$ -primary part annihilates exactly the monomials whose exponent is $\equiv -1 \pmod{p^v}$. The surviving exponents form the signature set

$$S = S(m, p^v) = \{k \in \{0, \dots, m - 1\} : k \bmod p^v \neq p^v - 1\}. \quad (1)$$

The *within-pair survivor count* is the number of distinct exponent pairs the two A -variables of the block realize. Empirically it takes one of two values, historically called *free* and *coupled*:

$$\text{free}(m, p^v) = |S|^2, \quad \text{triangle}(m, p^v) = \#\{(a, b) \in S \times S : (b \bmod p^v) \leq (a \bmod p^v)\}. \quad (2)$$

The two measured anchor values from the reduction engine are $(4, 6)$ at $p = 3 \rightarrow 16$ (free) and $(4, 12)$ at $p = 2 \rightarrow 54$ (coupled).

2 The factorization and the quotient law

Lemma 1 (Residue-class structure of S). *S is the disjoint union of exactly $p^v - 1$ residue classes modulo p^v — namely the classes $0, 1, \dots, p^v - 2$ — each of cardinality $c = m/p^v$. The class $p^v - 1$ is absent (annihilated by the A -idempotent). In particular $|S| = (p^v - 1)c$.*

Proof. By (1), $k \in S$ iff $k \bmod p^v \in \{0, \dots, p^v - 2\}$. Each such residue is attained by exactly $m/p^v = c$ values of k in $\{0, \dots, m - 1\}$, since $p^v \mid m$. The residue $p^v - 1$ is excluded, leaving $p^v - 1$ classes of size c . $\square$

Theorem 1 (Within-pair factorization). *Write each surviving exponent by its residue class $r \in \{0, \dots, p^v - 2\}$ and its position within that class. The within-pair survivor count factorizes as a within-class factor times an across-class factor, with the within-class factor always equal to c^2 , and*

$$\text{free} = c^2 (p^v - 1)^2, \quad \text{triangle} = c^2 T(p^v - 1) = c^2 \cdot \frac{(p^v - 1)p^v}{2}, \quad (3)$$

where $T(k) = k(k + 1)/2$.

Proof. By Lemma 1 a pair $(a, b) \in S \times S$ is determined by the residue pair (r_a, r_b) and, independently, by the within-class positions, of which there are c for each of a, b — a factor c^2 carrying no constraint between a and b . For the free count every residue pair is allowed, giving $(p^v - 1)^2$ residue pairs and $\text{free} = c^2(p^v - 1)^2$. For the triangle count we require $r_b \leq r_a$ and split on equality:

$$r_b < r_a : \binom{p^v - 1}{2} \text{ residue pairs}; \quad r_b = r_a : (p^v - 1) \text{ residue pairs};$$

each contributing c^2 element-pairs. Hence

$$\text{triangle} = c^2 \left[\frac{(p^v - 1)(p^v - 2)}{2} + (p^v - 1) \right] = c^2(p^v - 1) \left[\frac{p^v - 2}{2} + 1 \right] = c^2 \cdot \frac{(p^v - 1)p^v}{2}. \quad \square$$

Corollary 1 (Quotient law).

$$\frac{\text{triangle}}{\text{free}} = \frac{c^2(p^v - 1)p^v/2}{c^2(p^v - 1)^2} = \frac{p^v}{2(p^v - 1)}.$$

The within-class factor c^2 cancels exactly because it is identical in both regimes; the quotient is a pure function of the thickness p^v .

Corollary 2 (Degeneration). *triangle = free if and only if $p^v = 2$.*

Proof. $p^v / (2(p^v - 1)) = 1 \iff p^v = 2(p^v - 1) \iff p^v = 2$; equivalently $(p^v - 1)^2 = (p^v - 1)p^v/2 \iff p^v - 1 = 1$, i.e. a single live class, where triangle and square coincide. In particular the degeneration is governed by $p^v = 2$, *not* by the parity of p : the cell $(4, 8)$ at $p = 2$ has $p^v = 8$ and is coupled ($T(7) = 28 \neq 49 = (7)^2$). $\square$

Remark 1. A previously entertained explanation — that the operator ρ of the underlying construction imposes the order $r_b \leq r_a$ on the support — is false: the support reached by ρ is the *full* square of live-class pairs in every cell measured (e.g. $(4, 18)$ at $p = 3$ reaches all 64 pairs while the triangle has 36). The triangle is therefore not a constraint on which monomials appear; it is a statement about which survive. This motivates the functorial reading of Section 3.

3 The functorial identification

Let $V = \mathbb{F}_p^{p^v-1}$ be the live-class space (one basis vector per live residue class) and $W = \mathbb{F}_p^c$ the within-class space.

Theorem 2 (Functorial identification). *The within-pair survivor count is the dimension of a functor of V , tensored with the free within-class factor:*

$$\text{free} = \dim(V \otimes V) \cdot c^2 = (p^v - 1)^2 c^2, \quad \text{coupled} = \dim(\text{Sym}^2 V) \cdot c^2 = T(p^v - 1) c^2.$$

The free/coupled distinction is the single statement that, on the class layer, the tensor square $V \otimes V$ is replaced by the symmetric square $\text{Sym}^2 V$. The triangular index set $\{(a, b) : b \leq a\}$ is exactly the canonical basis of $\text{Sym}^2 V$ (the size-two multisets), which is why the triangle is what survives a full-square support.

Proof. Immediate from Theorem 1: $\dim(V \otimes V) = (p^v - 1)^2$ and $\dim(\text{Sym}^2 V) = \binom{p^v-1+1}{2} = T(p^v - 1)$, and the within-class factor $W \otimes W$ has dimension c^2 in both regimes. The identification of the triangular index set with the multiset basis of $\text{Sym}^2 V$ is the standard basis of the symmetric square. $\square$

Corollary 3. *The degeneration $p^v = 2$ is the geometric fact $\text{Sym}^2(\mathbb{F}_p^1) = \mathbb{F}_p^1 = \mathbb{F}_p^1 \otimes \mathbb{F}_p^1$: the symmetric square of a line is the line. The decomposition*

$$V \otimes V = \text{Sym}^2 V \oplus \Lambda^2 V, \quad (p^v - 1)^2 = \frac{(p^v - 1)p^v}{2} + \frac{(p^v - 1)(p^v - 2)}{2},$$

holds over $\mathbb{Z}$ and over $\mathbb{F}_p$ for p odd; the coupling keeps $\text{Sym}^2 V$ and discards $\Lambda^2 V$.

4 The realization question, and the characteristic-two obstruction

Theorem 2 identifies the coupled survivor with $\dim \text{Sym}^2 V$. It remains to prove that the relevant symmetrization actually *realizes* $\text{Sym}^2 V$ — that its image is all of $\text{Sym}^2 V$, not a proper subspace. The symmetrization in play is the swap τ of the two pair variables: interchanging the two A -variables is precisely τ acting on $V \otimes V$, and ρ contributes only *which* classes are live (the set S), not a separate symmetrizer. The realization question is therefore the surjectivity of $I + \tau : V \otimes V \rightarrow V \otimes V$ onto $\text{Sym}^2 V$, equivalently the cokernel of the symmetrization.

Theorem 3 (Realization). *Let $\sigma = I + \tau$ on $V \otimes V$, $V = \mathbb{F}_p^{p^v-1}$.*

- (i) *For p odd, $\text{im}(\sigma) = \text{Sym}^2 V$ over $\mathbb{F}_p$: the symmetrization is surjective onto the symmetric square, and the coupled identity $\text{coupled} = \dim(\text{Sym}^2 V) \cdot c^2$ holds as an $\mathbb{F}_p$ -statement.*
- (ii) *For $p = 2$, $\text{im}(\sigma)$ falls short of $\text{Sym}^2 V$ by exactly $p^v - 1$ dimensions; the integral cokernel of σ carries torsion $(\mathbb{Z}/2)^{p^v-1}$, concentrated on the diagonal.*

Proof (computational, byte-exact). Over $\mathbb{Z}$ the matrix of $\sigma = I + \tau$ in the basis $e_i \otimes e_j$ sends $e_i \otimes e_j \mapsto e_i \otimes e_j + e_j \otimes e_i$; off-diagonal symmetric vectors receive multiplicity one and the diagonal vectors $e_i \otimes e_i$ receive multiplicity two. The Smith normal form over $\mathbb{Z}$ has, among its nonzero elementary divisors, exactly $p^v - 1$ entries equal to 2 (one per diagonal class) and the rest units. Reducing modulo p : for p odd the 2's are units, so the $\mathbb{F}_p$ -rank equals the $\mathbb{Z}$ -rank and $\text{im}(\sigma) = \text{Sym}^2 V$; for $p = 2$ the $p^v - 1$ diagonal entries are divisible by p , so the $\mathbb{F}_p$ -rank drops by $p^v - 1$ and the cokernel acquires $(\mathbb{Z}/2)^{p^v-1}$. This is verified byte-exact by Smith normal form and, independently, by the rank drop of σ over $\mathbb{F}_p$ versus , across the cells listed in Section 6. $\square$

Theorem 4 (The obstruction is generic). *The cokernel torsion $(\mathbb{Z}/2)^{p^v-1}$ of Theorem 3(ii) depends only on $\dim V = p^v - 1$ and not on the Fermat data: for any $\mathbb{F}_p$ -space V of dimension n , the cokernel of $I + \tau$ over $\mathbb{Z}$ carries torsion $(\mathbb{Z}/2)^n$ in characteristic two, the diagonal where the symmetric and alternating squares coincide modulo 2. The Fermat construction inherits this torsion exactly — it adds and removes not a single factor of 2.*

Proof. The argument of Theorem 3 uses only the swap τ on $V \otimes V$ and the multiplicity-two diagonal; no Fermat input enters. Hence the count is $n = \dim V$ for a generic V , and ρ enters only through $\dim V = p^v - 1$. Verified byte-exact for plain V of dimensions 1, 2, 3, 4, 7. $\square$

Corollary 4 (The two threads are one). *The realization question (is σ surjective onto $\mathrm{Sym}^2 V$?) and the torsion question (does characteristic two carry torsion in the Sym^2/Λ^2 layer?) are the same question: σ is surjective $\iff \mathrm{coker}(\sigma) = 0 \iff p$ odd, and in characteristic two $\mathrm{coker}(\sigma) = (\mathbb{Z}/2)^{p^v-1}$ is the generic swap torsion. The two converge byte-exact.*

5 Significance and scope

What is proved. The within-pair survivor count is the symmetric square $\mathrm{Sym}^2 V$ tensored with the free within-class factor (Theorem 2); the realization $\mathrm{im}(\sigma) = \mathrm{Sym}^2 V$ is a theorem in odd characteristic (Theorem 3(i)); and the characteristic-two shortfall is the *generic* swap torsion $(\mathbb{Z}/2)^{p^v-1}$, not a Fermat-specific phenomenon (Theorems 3(ii), 4).

What this closes for the swan hunt. The within-pair class layer is closed as a site for a Fermat-specific integral torsion counterexample: its only characteristic-two torsion is the universal swap torsion, identical for Fermat and for a generic space of the same dimension. This is the closure of one candidate layer with data, by Smith normal form — the correct tool, since torsion is detected by elementary divisors and rank drop, never by equality of $\mathbb{F}_p$ - and -dimensions of a subspace.

What this does not give. The result concerns one layer (the within-pair class layer). Other layers — the full block, the B -factor, the cross-pair structure, and the high-dimensional ambient module — are not addressed and remain open. The coupled identity is a theorem in odd characteristic; in characteristic two the obstruction is identified and shown generic, but the coupled identity is not an $\mathbb{F}_p$ -equality there. No claim is made on the rational (Clay) Hodge conjecture; this concerns the integral Hodge conjecture for Fermat varieties, a specific bounded problem. The torsion counterexample has not been found; every measured cell is primitive.

6 Reproducibility

Every numerical statement above is recomputed from scratch by a self-contained verifier. The quotient law and functorial identification are checked over the cells

$$(4, 6), (4, 8), (4, 12), (4, 9), (4, 18), (4, 25), (4, 14), (4, 16), (4, 21)$$

at the relevant primes; the realization theorem and the generic-torsion theorem are checked by Smith normal form over $\mathbb{Z}$ and by the $\mathbb{F}_p$ -versus- rank drop of $I + \tau$, in each cell's own characteristic, for $p \in \{2, 3, 5, 7\}$. The diagonal multiplicity-two structure and the $(\mathbb{Z}/2)^{p^v-1}$ cokernel are confirmed in characteristic two and shown absent in odd characteristic. The checks return zero discrepancies.

References

- [1] R. Amichis Luengo, *The localization of torsion to a single CRT block*. Campaign note, version 1.1, 1 June 2026.
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- [3] E. Aljovin, H. Movasati, R. Villafior, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628.